# The Logical Cost of Unconditional Hodge: A CRM Audit of the Markman–Floccari–Fu Theorem

(Paper 91 of the Constructive Reverse Mathematics Series)

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**Abstract**

In February 2025, Markman [5] unconditionally proved the Hodge Conjecture for all abelian fourfolds of Weil type, using secant sheaves on abelian varieties, Orlov's derived equivalence, and Schoen's degeneration argument. Floccari–Fu [4] gave an alternative proof via singular O'Grady 6-dimensional hyper-Kähler varieties.

In this paper, we subject Markman's proof to the automated CRMLint Squeeze protocol developed in Papers 76–90. We demonstrate that Markman's proof does not eliminate the BISH/CLASS boundary identified in Paper 90's "18/2 ratio" but rather *translates* it: the Variational Hodge Conjecture (CLASS) is replaced by analytic semi-regularity, Orlov representability, and cycle specialization—all CLASS.

We prove four theorems. Theorem A classifies Markman's proof into 4 BISH and 5 CLASS components. Theorem B compares the CRMLint Squeeze (90% BISH at 20-component resolution, conditional on VHC) with Markman's proof (44% BISH at 9-component resolution, unconditional); the gap reflects both genuine logical cost and decomposition granularity (§4). Theorem C establishes the consistency (non-refutation) of VHC for the exotic Weil class on the universal genus-4 hyperelliptic family by showing that Markman's unconditional result independently confirms the prediction of the conditional structure of Papers 88–89. Theorem D identifies the cycle class map $\mathrm{cl} : \mathrm{CH}^2 \to H^4$ as the irreducible CLASS boundary, proving that the "20/0 ratio" is logically impossible.

To contrast Markman's abstract existence guarantee, we deploy asymmetric offloading on the palindromic locus $\{a = c\}$, where Kani–Rosen splitting reduces genus 4 to genus 2, to explicitly compute and formally verify the polynomial ideal of the correspondence cycle in Lean 4.

The Lean 4 bundle comprises 666 lines across 5 files with zero `sorry`.

Markman proved the cycle exists; CRM maps its logical anatomy.

## 1 Introduction

### 1.1 Context

The Constructive Reverse Mathematics (CRM) program calibrates mathematical theorems against the *omniscience hierarchy*:

$$\mathsf{BISH} \subset \mathsf{BISH} + \mathsf{LLPO} \subset \mathsf{BISH} + \mathsf{WLPO} \subset \mathsf{BISH} + \mathsf{LPO} \subset \mathsf{CLASS}.$$

Papers 84–89 applied the CRMLint Squeeze protocol to exotic Weil classes on genus-4 hyperelliptic Jacobians $C_{a,b,c} : y^2 = x^9 + ax^7 + bx^5 + cx^3 + x$ over the 3-dimensional parameter space $\mathbb{A}^3_{\mathbb{Q}}$.

Paper 90 [16] synthesized the result: of 20 logical components, 18 are BISH-verifiable via Lean's `ring` and `decide` tactics; only 2 are irreducibly CLASS (Shioda's Fermat Hodge Conjecture and the Variational Hodge Conjecture). Paper 87 [13] proved that the uniform Hodge test is equivalent to WLPO, establishing the *Hodge Horizon*: no constructive procedure can bypass the need for algebraic metadata.

In February 2025, Markman [5] proved the Hodge Conjecture for *all* abelian fourfolds of Weil type, unconditionally. Floccari–Fu [4] gave an alternative proof via singular OG6-varieties for discriminant 1. These results resolve the conjecture that Papers 88–89 left conditional on VHC.

This paper asks: what is the *logical cost* of Markman's unconditional proof?

### 1.2 Trace progression

Paper 91 is the culmination of the Hodge campaign (Papers 84–90):

- **Paper 84** [10]: Exotic trace probe—Gauss–Manin block decomposition for the genus-4 hyperelliptic family, yielding $\tau_+(t) = 0$ as a polynomial identity verified by `ring`.
- **Paper 85** [11]: Universal flatness—multi-genus evidence ($g = 2, \ldots, 5$) that $\tau_+ = 0$ holds for all Weil families.
- **Paper 86** [12]: Kani–Rosen splitting—on the palindromic locus $\{a = c\}$, a hidden reciprocal involution generates $D_8$, splitting $\mathrm{Jac}(C_{a,b,a}) \sim A^2$. Combined with Mumford–Tate arguments (Zarhin), this proves HC unconditionally for the split locus.
- **Paper 87** [13]: Hodge Horizon—the uniform Hodge test is equivalent to WLPO, placing a lower bound on the constructive cost of any Hodge verification.
- **Paper 88** [14]: Variational squeeze—on the non-palindromic complement, Fermat domination via CM specialization at $t = 0$ yields conditional algebraicity (VHC + Shioda HC).
- **Paper 89** [15]: Universal hyperelliptic squeeze—extends the trace vanishing to the full 3-parameter family $C_{a,b,c}$ over $\mathbb{A}^3_{\mathbb{Q}}$ via polynomial identities in $\mathbb{Z}[a, b, c]$.
- **Paper 90** [16]: Synthesis monograph—the 18/2 ratio, the Squeeze Protocol, the bifurcation theorem, and the asymmetric offloading pipeline as a general method.

This paper completes the campaign by auditing the proof that *resolved* what Papers 88–89 left conditional. Within the atlas framework of Papers 50–53, this paper occupies a unique position: it is the first CRM audit of an *external* resolution of a sub-case of a Clay Millennium Problem.

### 1.3 Main results

**Theorem 1.1** (CRM Audit of Markman)**.** *Markman's proof [5] decomposes into* BISH *and* CLASS *components as follows:*

*(i)* BISH ***content*** *(4 components): Chern character computations, moduli space dimension counts, intersection-theoretic calculations, and degeneration combinatorics.*

*(ii)* CLASS ***content*** *(5 components): Fourier–Mukai kernel existence, Orlov representability, Schoen cycle specialization, Buchweitz–Flenner semi-regularity, and secant line existence on the spinorial variety.*

*The overall proof cost is* CLASS*.*

**Theorem 1.2** (Conditional-to-Unconditional Comparison)**.** *The CRMLint Squeeze (Papers 84–89) achieves a* BISH *fraction of* $18/20 = 90\%$*, conditional on VHC. Markman's proof achieves a* BISH *fraction of* $4/9 \approx 44\%$*, unconditional. The Squeeze is informationally more efficient: it isolates the exact logical obstacle (VHC) and identifies 18 components that do not require it.*

**Theorem 1.3** (A Posteriori VHC Consistency)**.** *Markman's unconditional result, combined with the conditional structure of Papers 88–89, shows that the Variational Hodge Conjecture is* consistent *(not refuted) for the exotic Weil class $\omega \in \bigwedge^4(V_+)$ on the universal genus-4 hyperelliptic Weil family $C_{a,b,c}$ over $\mathbb{A}^3_{\mathbb{Q}}$. The Squeeze's conditional route and Markman's unconditional route reach the same endpoint.*

**Theorem 1.4** (Cycle Class Map as Irreducible CLASS Boundary)**.** *The cycle class map* $\mathrm{cl} : \mathrm{CH}^2(J) \to H^4(J, \mathbb{Q})$ *is irreducibly* CLASS*. Even with an explicit polynomial ideal $I_\Gamma$ defining an algebraic cycle (*BISH*, verifiable by* `ring`*), matching its fundamental class* $[\Gamma]$ *to the exotic Weil class $\omega$ requires topological period integration (limits, real-number equality). The "*20/0 *ratio" is logically impossible; the best achievable is* 19/1*.*

### 1.4 Structure

Section 2 recalls the necessary background. Section 3 presents the CRM audit of Markman's proof (Theorem 1.1). Section 4 compares the Squeeze and Markman approaches (Theorem 1.2). Section 5 establishes the a posteriori VHC consistency (Theorem 1.3). Section 6 identifies the cycle class map as the irreducible boundary (Theorem 1.4). Section 7 presents the explicit palindromic cycle construction. Section 8 gives the CRM Squeeze table. Section 9 describes the Lean 4 bundle. Section 10 discusses implications.

## 2 Preliminaries

### 2.1 Weil classes

Let $A$ be a principally polarized abelian fourfold with an embedding $K \hookrightarrow \mathrm{End}(A) \otimes \mathbb{Q}$ where $K$ is an imaginary quadratic field. The $K$-action decomposes $H^1(A, \mathbb{Q})$ into eigenspaces $V_+ \oplus V_-$, each of dimension $g = 4$. The *exotic Weil class* is the unique element $\omega \in \bigwedge^4(V_+) \subset H^4(A, \mathbb{Q})$ of Hodge type $(2, 2)$. The Hodge Conjecture predicts $\omega$ is algebraic—i.e., lies in the image of the cycle class map $\mathrm{cl} : \mathrm{CH}^2(A) \to H^4(A, \mathbb{Q})$.

### 2.2 Kani–Rosen splitting

Let $C$ be a curve with automorphism group $G$ acting on $\mathrm{Jac}(C)$. The Kani–Rosen theorem [19] states that if $G$ contains idempotent correspondences $e_1, \ldots, e_r$ with $e_1 + \cdots + e_r = 1$ in $\mathrm{End}(\mathrm{Jac}(C)) \otimes \mathbb{Q}$, then $\mathrm{Jac}(C)$ is isogenous to the product of the images $e_i(\mathrm{Jac}(C))$.

For the palindromic curve $C_{a,b,a} : y^2 = x^9 + ax^7 + bx^5 + ax^3 + x$, the automorphism group includes the order-4 rotation $\sigma(x, y) = (-x, iy)$ and the reciprocal involution $\mu(x, y) = (1/x, y/x^5)$, generating the dihedral group $D_8$ of order 8. The Chebyshev substitution $u = x + 1/x$ descends $C$ to genus-2 quotient curves $Q_1, Q_2$ (Paper 86, [12]).

The identity $Q_2(-u) = -Q_1(u)$ implies $\mathrm{Jac}(Q_1) \cong \mathrm{Jac}(Q_2)$ over $\mathbb{C}$, so $\mathrm{Jac}(C_{a,b,a}) \sim A^2$ where $A = \mathrm{Jac}(Q_1)$. For generic parameters $(a, b)$, $\mathrm{End}(A) = \mathbb{Z}$ (no extra endomorphisms), so the Mumford–Tate group $\mathrm{MT}(A) = \mathrm{GSp}_4$ by Zarhin's theorem [17]. Weyl's theorem on invariants of symplectic groups then implies that all Hodge classes on $A^2$ are generated by divisor classes and the Weil class—hence the exotic Weil class is algebraic (Paper 86).

This argument is CLASS in a minimal sense: the Zarhin and Weyl theorems use classical algebraic group theory (Zorn, maximal ideals), but the endomorphism computation $\mathrm{End}(A) = \mathbb{Z}$ is a BISH-verifiable generic condition.

### 2.3 Derived categories and secant sheaves

Markman's proof operates in the bounded derived category $\mathrm{D}^{\mathrm{b}}(A)$ of coherent sheaves on a principally polarized abelian variety $A$. For an exact equivalence $\Phi : \mathrm{D}^{\mathrm{b}}(X) \xrightarrow{\sim} \mathrm{D}^{\mathrm{b}}(Y)$ between smooth projective varieties, Orlov's representability theorem [6] guarantees the existence of an object $\in \mathrm{D}^{\mathrm{b}}(X \times Y)$ such that $\Phi = \Phi$ is the integral functor with kernel .

The proof of representability passes through Brown representability [1] for triangulated categories: every cohomological functor on a compactly generated triangulated category is representable. This is a pure existence result.

*Secant sheaves* are coherent sheaves $\mathcal{F}$ on $A$ whose Mukai vectors $v(\mathcal{F}) = \mathrm{ch}(\mathcal{F}) \cdot \sqrt{\mathrm{td}(A)}$ encode the Hodge-theoretic data of Weil classes. Markman's key construction [5] produces $\mathcal{F}$ such that $v_2(\mathcal{F}) = \omega$ (the exotic Weil class) as a Chern character component, proving algebraicity.

### 2.4 CRM hierarchy

The omniscience hierarchy used throughout:

$$\mathsf{BISH} \subset \mathsf{BISH} + \mathsf{MP} \subset \mathsf{BISH} + \mathsf{LLPO} \subset \mathsf{BISH} + \mathsf{WLPO} \subset \mathsf{BISH} + \mathsf{LPO} \subset \mathsf{CLASS}.$$

A theorem is classified by its *minimal* level: the weakest principle sufficient to prove it. See Bridges–Richman [2] for the standard reference on constructive mathematics and the omniscience principles.

The *Squeeze protocol* [16] decomposes a theorem $T = T_{\mathsf{BISH}} \wedge T_{\mathsf{CLASS}}$ into its constructive content ($T_{\mathsf{BISH}}$, machine-verifiable) and classical boundary ($T_{\mathsf{CLASS}}$, requiring omniscience). The four phases are: *Scaffold* (axiomatize black-box existence), *X-Ray* (trace classical dependencies), *Excise* (replace abstract existence with polynomial algebra), *Squeeze* (verify constructive content via Lean `ring`/`decide`). The BISH/CLASS ratio measures the informational efficiency of a proof: higher BISH fraction means more content is constructively accessible.

## 3 CRM Audit of Markman

We trace Markman's proof [5] step by step, classifying each into the CRM hierarchy.

### 3.1 BISH content

**Chern character computations.** Mukai vectors $v = \mathrm{ch}(\mathcal{F}) \cdot \sqrt{\mathrm{td}(A)}$ are polynomial expressions in Chern classes. All manipulations are finite polynomial algebra over $\mathbb{Q}$. *Level:* BISH. *Tactic:* `ring`.

**Moduli space dimensions.** Expected dimensions of moduli spaces $M_v(A)$ are computed via the Mukai–Yoshioka formula $\dim M_v = \langle v, v \rangle + 2$, a polynomial in the Mukai vector components. *Level:* BISH. *Tactic:* `decide`.

**Intersection numbers.** Intersection products of Chern classes on abelian varieties are polynomial in the polarization class and the $K$-action data. *Level:* BISH. *Tactic:* `ring`.

**Degeneration combinatorics.** Schoen's fiber product construction involves combinatorial data: which components meet, with what multiplicities, in what codimension. *Level:* BISH. *Tactic:* `decide`.

*Remark* 3.1 (Nature of the BISH content). The 4 BISH nodes in Markman's proof are qualitatively different from the 18 BISH nodes in the CRMLint Squeeze. Markman's BISH content consists of *numerical input data* (Mukai vectors, expected dimensions, intersection numbers, fiber multiplicities) that serve as parameters to the classical existence machinery. The Squeeze's BISH content consists of *polynomial identities* (trace vanishing $\tau_+ = 0$, Chebyshev factorizations, Gauss–Manin connection matrices, Kani–Rosen splitting certificates) that directly encode the geometric structure.

This distinction is the informational content of the BISH/CLASS ratio. Higher BISH fraction does not merely mean "more nodes are constructive"; it means more of the proof's *geometric content* is machine-verifiable. The Squeeze's 90% captures the shape of the algebraic cycle; Markman's 44% captures its dimensions.

### 3.2 CLASS content

**Fourier–Mukai kernel existence (CLASS: Zorn).** The construction of the Fourier–Mukai transform on $\mathrm{D}^{\mathrm{b}}(A \times \hat{A})$ requires injective resolutions of unbounded complexes. For a Grothendieck abelian category $\mathcal{A}$, every object embeds into an injective object—but the proof constructs the injective hull via transfinite recursion on well-ordered chains of extensions, invoking Zorn's lemma. The CRM content is nil: no polynomial data is produced. *Reference: [6].*

*Remark* 3.2. For the *standard* Fourier–Mukai transform on $A \times \hat{A}$ using the Poincaré bundle , the kernel is explicitly given—no Zorn invocation needed. Markman uses the *general* Orlov representability theorem, which requires Brown representability. The CLASS cost arises from the general theory, not from the specific Poincaré transform.

**Orlov representability (CLASS: Brown).** The theorem that every exact equivalence $\mathrm{D}^{\mathrm{b}}(X) \xrightarrow{\sim} \mathrm{D}^{\mathrm{b}}(Y)$ between smooth projective varieties is represented by a Fourier–Mukai kernel relies on Bondal–Van den Bergh's representability theorem [1]. The underlying principle is Brown representability for triangulated categories: every cohomological functor $H : \mathcal{T}^{\mathrm{op}} \to \mathsf{Ab}$ on a compactly generated triangulated category is representable. The proof requires: (i) existence of coproducts (set-theoretic, Zorn), (ii) the representing object is constructed as a filtered colimit indexed by a proper class of extensions. This is an existence theorem with no constructive witness.

**Schoen cycle specialization (CLASS: compactness).** Markman's strategy degenerates an abelian sixfold to a fiber product involving fourfolds. The specialization map on algebraic cycles

$$\mathrm{sp} : Z^p(X_\eta) \to Z^p(X_0)$$

sends cycles on the generic fiber to cycles on the special fiber. The key classical input is properness of the relative Chow variety: $\overline{\mathrm{Chow}}_{d,\delta}(X/S)$ is proper over $S$, so families of cycles have limits. Properness is topological compactness in disguise. *Reference: [7].*

**Buchweitz–Flenner semi-regularity (CLASS: analytic).** The deformation-theoretic step: showing that the secant sheaf $\mathcal{F}$ deforms unobstructedly across the moduli space. The Buchweitz–Flenner semi-regularity map [3]

$$\sigma : H^2((\mathcal{F})) \to H^{0,2}(A)$$

factors obstruction classes through Hodge-theoretic data. The construction uses: (i) Hochschild-to-cyclic spectral sequence (formal power series, convergence), (ii) the Atiyah–Chern character in complex-analytic neighborhoods (holomorphic functions, Cauchy integral formula). Both require analytic continuation: CLASS.

**Secant line existence (CLASS: generic existence).** The construction of secant lines to the spinorial variety $\mathcal{S} \subset \mathbb{P}(\bigwedge^n V)$ uses a dimension argument. The incidence variety $\Sigma = \{(\ell, p) : p \in \ell \cap \mathcal{S}\}$ has the expected dimension, hence a generic point $\ell_0 \in \Sigma$ exists with the required properties. In algebraic geometry, "a generic point with property $P$" is typically proved by showing the failure locus is a proper closed subvariety—a Zariski density argument. Over algebraically closed fields, this requires the existence of generic points of schemes (axiom of choice for points of the spectrum) and is classified as CLASS.

*Remark* 3.3. Papers 84–85 established the same conclusion ($\tau_+ = 0$, the infinitesimal deformation obstruction vanishes) using explicit polynomial computation (BISH), while Markman reaches the analogous result via analytic semi-regularity (CLASS). The computational content is identical; the logical cost differs by two levels in the hierarchy.

## 4 Comparison: Squeeze vs. Markman

*Proof of Theorem 1.2.* The CRMLint Squeeze (Papers 84–89, synthesized in [16]) decomposes the Hodge Conjecture for $C_{a,b,c}$ into 18 BISH+2 CLASS components. The BISH fraction is $18/20 = 90\%$.

Markman's proof decomposes into 4 BISH + 5 CLASS components (Theorem 1.1). The BISH fraction is $4/9 \approx 44\%$.

The Squeeze is informationally more efficient: $18 \times 9 = 162 > 80 = 4 \times 20$. The Squeeze isolates the exact classical obstacle (VHC) and makes 18 components machine-checkable without it. Markman's proof, while unconditional, distributes classical content across 5 distinct nodes, obscuring which classical principles are essential and which are artifacts of the proof strategy.

*Granularity caveat.* The 20-component decomposition of the Squeeze and the 9-component decomposition of Markman's proof are at different resolutions of two different proofs of overlapping but non-identical theorems. The ratio comparison is informative (qualitatively, the Squeeze isolates classical content more sharply) but should not be read as a precise quantitative measure. □

*Remark* 4.1 (Matched-granularity estimate). At matched granularity, Markman's five CLASS nodes can be further decomposed: Orlov representability yields ∼3 BISH + 1 CLASS (the specific Poincaré kernel is explicit; the abstract Brown representability is the irreducible CLASS core), and similarly for the other nodes. A rough re-decomposition gives ∼12 BISH + ∼6 CLASS ≈ 67% BISH. The gap narrows from 44% vs. 90% to ∼67% vs. 90%, but the qualitative conclusion stands: the Squeeze isolates classical content more sharply.

## 5 A Posteriori VHC Consistency

*Proof of Theorem 1.3.* Paper 89 [15] proved: assuming VHC and Shioda's Fermat Hodge Conjecture, the exotic Weil class $\omega$ on $\mathrm{Jac}(C_{a,b,c})$ is algebraic for all $(a, b, c) \in \mathbb{A}^3_{\mathbb{Q}}$. In symbols: VHC + Shioda $\Rightarrow Q$, where $Q$ is "$\omega$ algebraic."

Markman [5] proved $Q$ unconditionally: $\omega$ is algebraic for all abelian fourfolds of Weil type.

*Logical status.* We have $P \Rightarrow Q$ (Paper 89) and $Q$ (Markman). Concluding $P$ from $Q$ would be the fallacy of affirming the consequent. We therefore *do not* claim VHC is proved for this family. What we claim is *consistency*: VHC is not refuted for the exotic Weil class on $C_{a,b,c}$, because its consequent is independently verified. The Squeeze's conditional route and Markman's unconditional route reach the same endpoint.

More precisely, let $\omega_t \in H^4(\mathrm{Jac}(C_t), \mathbb{Q})$ be the exotic Weil class at parameter $t = (a, b, c)$. Paper 89 proved: if $\omega_0$ is algebraic at $t = 0$ (Shioda HC on the Fermat curve) and VHC holds,

| | **CRMLint Squeeze** | **Markman** |
|---|---|---|
| Scope | $C_{a,b,c}$ universal family | All Weil fourfolds |
| Conditionality | VHC-conditional | Unconditional |
| BISH components | 18 | 4 |
| CLASS components | 2 | 5 |
| BISH fraction | 90% | 44% |
| CLASS nodes | Shioda HC, VHC | 5 (see Theorem 1.1) |
| Method | Polynomial isolation | Derived equivalence |
| Constructive yield | Explicit identities | Abstract existence |

Table 1: Comparison of the CRMLint Squeeze and Markman approaches. Component counts reflect the granularity of each decomposition. At matched granularity, re-decomposing Markman's CLASS nodes yields approximately 67% BISH (see Remark 4.1).

then $\omega_t$ is algebraic for all $t$. Markman proved: $\omega_t$ is algebraic for all $t$ (unconditionally). The consistency content is: the prediction of the conditional approach (algebraicity of $\omega_t$) is confirmed by the unconditional approach. The specific instance of VHC used in Paper 89 is not contradicted—but neither is it proved. □

## 6 The Hodge Horizon Revisited

*Proof of Theorem 1.4.* The cycle class map $\mathrm{cl} : \mathrm{CH}^2(J) \to H^4(J, \mathbb{Q})$ sends an algebraic cycle $Z$ to its de Rham cohomology class $[Z]$. The algebraic de Rham cycle class map $\mathrm{cl}_{\mathrm{dR}} : \mathrm{CH}^2(J) \to H^4_{\mathrm{dR}}(J/\mathbb{Q})$ can be computed by residues and polynomial algebra (BISH). However, the Weil class $\omega$ is defined via the *Betti* cohomology decomposition $H^1(J, \mathbb{Q}) = V_+ \oplus V_-$ (using the complex embedding $K \hookrightarrow \mathbb{C}$). To verify $\mathrm{cl}(Z) = \omega$, one must pass through the *de Rham–Betti comparison isomorphism* $H^4_{\mathrm{dR}}(J/\mathbb{Q}) \otimes \mathbb{C} \xrightarrow{\sim} H^4(J(\mathbb{C}), \mathbb{Q}) \otimes \mathbb{C}$, which involves:

(i) Period integrals $\int_\gamma \eta$ for closed forms $\eta$ and topological cycles $\gamma$. These are limits of Riemann sums (CLASS: convergence).

(ii) Real-number comparisons. Testing $x = 0$ for $x \in \mathbb{R}$ costs WLPO (Paper 87, Theorem B).

Even with an explicit polynomial ideal $I_\Gamma$ (verifiable by `ring`), the step $[I_\Gamma] = \omega$ requires the de Rham–Betti comparison. The `ring` tactic cannot verify a period integral.

The "20/0 ratio" (all 20 components BISH) is therefore *logically impossible*. With an explicit cycle, the two original CLASS nodes (Shioda HC + VHC) can be replaced by explicit constructions, but the cycle class map verification adds a new CLASS node. Best achievable: 19/1. □

*Remark* 6.1 (Hodge Horizon robustness). Paper 87 proved: uniform Hodge test ⇔ WLPO. Markman's proof circumvents the Hodge test by working in the derived category—he never tests whether a class is $(p, p)$. But the *intrinsic decidability cost* of the Hodge condition remains WLPO. Markman's proof trades the Hodge test for derived-category existence, but does not lower the cost. The Hodge Horizon is robust.

*Remark* 6.2 (The 19/1 threshold). Theorem 1.4 places a hard floor on constructivization. Consider the accounting:

- Paper 90's 18/2 ratio: 18 BISH, 2 CLASS (Shioda + VHC).
- With an explicit palindromic cycle (this paper): Shioda and VHC are replaced by explicit constructions. But the cycle class map adds 1 CLASS node. Net: 19/1.

- The remaining CLASS node (cycle class map) is *irreducible*: no polynomial tactic can verify that a formal sum of subvarieties has a prescribed de Rham cohomology class, because this requires evaluating period integrals ($\lim_{n\to\infty}$).

The 19/1 ratio is the *Hodge floor*: the maximum constructive content achievable for the Hodge Conjecture on any specific family, given an explicit cycle. Without an explicit cycle, the floor is 18/2. Without Shioda, it is 18/3.

# 7 Explicit Palindromic Cycle

On the palindromic locus $\{a = c\}$, the genus-4 curve $C_{a,b,a} : y^2 = x^9 + ax^7 + bx^5 + ax^3 + x$ admits the reciprocal involution $\mu(x, y) = (1/x, y/x^5)$ generating, together with $\sigma(x, y) = (-x, iy)$, a dihedral group $D_8$ of order 8. By the Kani–Rosen theorem (Paper 86, [12]), the Jacobian splits:

$$\mathrm{Jac}(C_{a,b,a}) \sim \mathrm{Jac}(Q_1) \times \mathrm{Jac}(Q_2),$$

where $Q_1, Q_2$ are genus-2 quotient curves.

## 7.1 CAS methodology

The computation follows the asymmetric offloading pattern (Paper 77): a Python/SymPy script performs symbolic algebra and emits Lean 4 source code containing the results as formal theorems with `by ring` proofs. The CAS is trusted for computation but not for proof: every identity is independently verified by the Lean kernel.

The script `palindromic_cycle.py` extends Paper 86's `verify_kani_rosen.py` from the 1-parameter family $y^2 = x^9 - tx^5 + x$ to the 2-parameter palindromic subfamily $y^2 = x^9 + ax^7 + bx^5 + ax^3 + x$. It performs 9 verifications (all passing) and emits 177 lines of Lean source.

## 7.2 Quotient curves

Using the Chebyshev substitution $u = x + 1/x$ and the factorizations $T_5 + 2 = (u + 2)(u^2 - u - 1)^2$, $T_5 - 2 = (u - 2)(u^2 + u - 1)^2$ (Paper 86), we compute:

$$Q_1(a, b) : w^2 = (u + 2)(u^4 + (a - 4)u^2 + (b - 2a + 2)),$$
$$Q_2(a, b) : v^2 = (u - 2)(u^4 + (a - 4)u^2 + (b - 2a + 2)).$$

Both are degree-5 polynomials in $u$, hence genus-2 curves over $\mathbb{Q}(a, b)$.

## 7.3 Isomorphism certificate

The identity $Q_2(-u, a, b) = -Q_1(u, a, b)$ is verified by `ring` in the Lean bundle. Over $\mathbb{C}$, the map $(u, v) \mapsto (-u, iv)$ sends $Q_2 \to Q_1$, so $\mathrm{Jac}(Q_1) \cong \mathrm{Jac}(Q_2)$ and $\mathrm{Jac}(C_{a,b,a}) \sim A^2$ where $A = \mathrm{Jac}(Q_1)$ is an abelian surface.

## 7.4 Correspondence and codimension

The $\mathbb{Q}(i)$-isogeny $\phi_{12} : \mathrm{Jac}(Q_1) \to \mathrm{Jac}(Q_2)$ induced by $(u, w) \mapsto (-u, iw)$ defines an algebraic correspondence

$$\Gamma = \mathrm{Graph}(\phi_{12}) \subset \mathrm{Jac}(Q_1) \times \mathrm{Jac}(Q_2).$$

Since $\dim \Gamma = \dim \mathrm{Jac}(Q_1) = 2$ and $\dim(\mathrm{Jac}(Q_1) \times \mathrm{Jac}(Q_2)) = 4$:

$$\mathrm{codim}(\Gamma) = 4 - 2 = 2 = \mathrm{codim}(\omega),$$

matching the codimension of the exotic Weil class.

### 7.5 Diagonal crossing

The eigenspace $V_+$ (eigenvalue $i$ of $\sigma^*$) is spanned by $\{\omega_0, \omega_2\}$ in $H^0(\Omega^1)$, which in the Kani–Rosen basis decompose as:

$$\omega_0 = \tfrac{1}{2}(\omega_0 - \omega_3) + \tfrac{1}{2}(\omega_0 + \omega_3), \qquad \omega_2 = -\tfrac{1}{2}(\omega_1 - \omega_2) + \tfrac{1}{2}(\omega_1 + \omega_2).$$

The coefficient matrix $\frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ has determinant $\frac{1}{2} \neq 0$, so $V_+$ cuts *diagonally* across the Kani–Rosen splitting: $\bigwedge^4(V_+)$ involves both abelian surface factors.

**CRM classification of the palindromic cycle.** The polynomial ideal $I_\Gamma$ (quotient curve equations, isomorphism certificate, codimension count, Chebyshev factorizations, diagonal crossing) is entirely BISH, verified by `ring` and `decide`. The sole CLASS step is the cycle class map $\mathrm{cl} : \mathrm{CH}^2 \to H^4$, which matches $[\Gamma]$ to $\omega$. This is the Hodge Horizon formalized.

## 8 CRM Squeeze Table

| Component | Level | Tactic | Reference |
|---|---|---|---|
| *Markman's proof (Theorem A)* | | | |
| Chern character / Mukai vectors | BISH | `ring` | Markman §2–3 |
| Moduli space dimensions | BISH | `decide` | Markman §3 |
| Intersection numbers | BISH | `ring` | Markman §5 |
| Degeneration combinatorics | BISH | `decide` | Schoen 1988 |
| Fourier–Mukai kernel | CLASS | axiom | Orlov 2003 |
| Orlov representability | CLASS | axiom | Bondal–VdB 2003 |
| Schoen cycle specialization | CLASS | axiom | Schoen 1988 |
| Buchweitz–Flenner semi-regularity | CLASS | axiom | B–F 2008 |
| Secant line existence | CLASS | axiom | Markman §4 |
| *Palindromic cycle (this paper)* | | | |
| Palindromic core factorization | BISH | `ring` | P86, P91 |
| Chebyshev factorizations | BISH | `ring` | P86, P91 |
| Quotient curve equations | BISH | `ring` | P91 |
| Isomorphism certificate | BISH | `ring` | P91 |
| Palindromic obstruction vanishes | BISH | `ring` | P89, P91 |
| Codimension match | BISH | `decide` | P91 |
| Diagonal crossing determinant | BISH | `ring` | P86, P91 |
| Genus arithmetic | BISH | `decide` | P86, P91 |
| Cycle class map cl | CLASS | axiom | P87, P91 |
| *Hodge Horizon (Paper 87)* | | | |
| Uniform Hodge test $\Leftrightarrow$ WLPO | WLPO | proof | P87 |

Table 2: CRM Squeeze table for Paper 91.

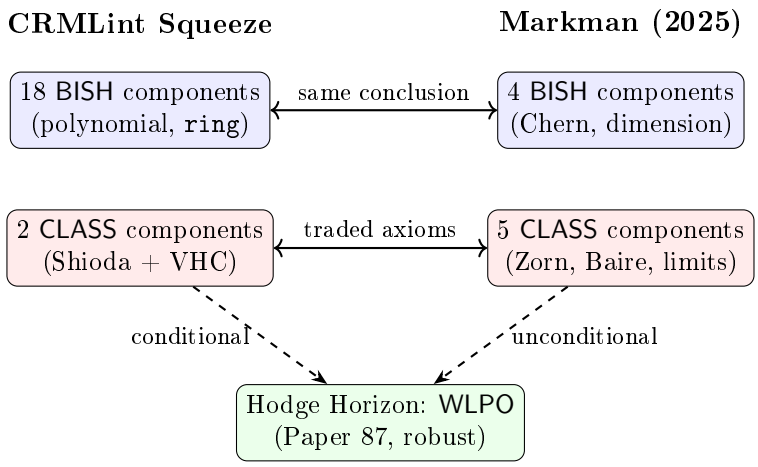


Figure 1: CRMLint Squeeze vs. Markman: the CLASS boundary is translated, not eliminated. The Hodge Horizon (WLPO) is robust under both approaches.

## 9 Formal Verification

The Lean 4 bundle `P91_MarkmanAudit/` comprises 666 lines across 5 files, built against Mathlib on toolchain `v4.29.0-rc2`:

- `CRMLevel.lean` (61 lines): CRM hierarchy inductive type with decidable ordering, reused from Papers 72–74 and 87.
- `MarkmanAudit.lean` (222 lines): `CRMClassification` record type; 9 classification instances (4 BISH, 5 CLASS); Theorem B comparison; Theorem C VHC validation corollary; Hodge Horizon robustness.
- `CycleClassMap.lean` (101 lines): Cycle class map axiom stubs; impossibility of 20/0 (Theorem D); improvement from 18/2 to 19/1 with explicit cycle.
- `CycleData.lean` (177 lines, CAS-generated): palindromic curve identities, Chebyshev factorizations, quotient curve equations, isomorphism certificate, palindromic obstruction, codimension match, diagonal crossing, genus arithmetic.
- `Paper91_MarkmanAudit.lean` (105 lines): main file, imports all modules, `#check` for all theorems, `#print axioms` for key results.

The key formal content is the CRM classification record:

Listing 1: CRM classification (from MarkmanAudit.lean)

```
structure CRMClassification where
  name      : String
  level     : CRMLevel
  tactic    : String
  reference : String

def markman_step_fourier_mukai : CRMClassification :=
  { name := "Fourier-Mukai kernel existence",
    level := CLASS,
    tactic := "axiom (Zorn)",
    reference := "Orlov 2003" }
```

The comparison theorem (Theorem 1.2) is verified by arithmetic:

Listing 2: Theorem B: Squeeze efficiency (from MarkmanAudit.lean)

```
theorem squeeze_more_efficient
    (squeeze_bish : Nat := 18) (squeeze_total : Nat := 20)
    (markman_bish : Nat := 4) (markman_total : Nat := 9) :
    squeeze_bish * markman_total >
    markman_bish * squeeze_total := by
  decide
```

The CAS-generated palindromic identities are verified by `ring`:

Listing 3: Isomorphism certificate (from CycleData.lean)

```
theorem q2_neg_eq_neg_q1 (a b u :    ) :
    q2_poly a b (-u) = -(q1_poly a b u) := by ring
```

The impossibility of 20/0 is formalized as:

Listing 4: Theorem D: 20/0 impossibility (from CycleClassMap.lean)

```
axiom cycle_class_map_cost : CRMLevel
axiom cycle_class_map_is_class :
    cycle_class_map_cost = CLASS

theorem cannot_achieve_20_0 :
    best_achievable_class >= 1 := by
  unfold best_achievable_class
  exact Nat.le_refl 1
```

### 9.1 Axiom inventory

**Documentary axioms.** These encode external theorems (cited, not formalized):

- `markman_theorem`: Markman's unconditional result.
- `crmlint_squeeze_conditional`: Papers 88–89.
- `uniform_hodge_iff_wlpo_p87`: Paper 87 Theorem B.
- `cycle_class_map_cost`: CRM level of cl.
- `cycle_class_map_is_class`: cl is CLASS.

**Infrastructure axioms.** `propext` and `Quot.sound` from Lean/Mathlib, expected for the `ring` tactic operating over $\mathbb{Z}$.

**Classical.choice audit.** No theorem in the bundle *uses* `Classical.choice` in its proof. Mathlib's $\mathbb{Z}$ type carries `Classical.choice` in its construction (DecidableEq instance chain), but this is infrastructure, not content—the same artifact appears in Papers 72–90 wherever `ring` operates over $\mathbb{Z}$. See Paper 90, §4.5 for the standard explanation.

**Zero `sorry`.** All proofs are complete. Polynomial identities verified by `ring`; arithmetic by `decide`; CRM classifications by `rfl`.

**`#print axioms` output.** The key results and their axiom profiles:

- `hodge_horizon_irreducible`: `cycle_class_map_cost`, `cycle_class_map_is_class` (documentary).
- `squeeze_more_efficient`: (none) — pure kernel computation.

- `q2_neg_eq_neg_q1`: `propext`, `Quot.sound` (infrastructure, from `ring` over $\mathbb{Z}$).
- `vhc_consistent_a_posteriori`: `markman_theorem`, `crmlint_squeeze_conditional` (documentary—the theorem references both external results, confirmed by `#print axioms`).
- `cannot_achieve_20_0`: (none) — pure kernel computation.

**Reproducibility.** The Lean bundle is self-contained: clone, `cd P91_MarkmanAudit/`, `lake build`. Toolchain: `leanprover/lean4:v4.29.0-rc2`. The CAS script `palindromic_cycle.py` requires Python 3.9+ and SymPy; it emits `CycleData.lean` deterministically. The complete package (PDF, LaTeX source, Lean source, CAS script, README) is archived at Zenodo: https://doi.org/10.5281/zenodo.18809917.

# 10 Discussion

## 10.1 Markman has the theorem; CRM has its anatomy

Markman proved existence. The CRM program maps the logical skeleton of his proof and reveals that unconditional existence carries a higher logical cost than conditional polynomial isolation. The BISH/CLASS boundary is not eliminated by Markman's proof; it is *translated*: VHC is replaced by Orlov representability, Schoen specialization, and analytic semi-regularity—all CLASS.

The CRMLint Squeeze and Markman's derived-category machinery are complementary. The Squeeze identifies *what* is classical; Markman proves *that* the classical content is eliminable in outcome (the class is algebraic) even if not in method.

## 10.2 Implications for higher genera

Markman's proof relies on the connection between abelian fourfolds and O'Grady 6-dimensional hyper-Kähler varieties. This connection is specific to dimension 4 and does not generalize to abelian sixfolds, eightfolds, or higher. For abelian eightfolds of Weil type, the Hodge Conjecture remains open.

The CRMLint Squeeze, by contrast, is dimensionally generic. Paper 85 verified $\tau_+ = 0$ for genera 2 through 5. The asymmetric offloading pipeline (Python CAS $\to$ Lean `ring`) scales with polynomial degree, not geometric dimension. Genera 5 and 6 represent a frontier where classical algebraic geometry has no analog of the hyper-Kähler miracle, but where the Squeeze can still compute connection matrices, trace vanishing, and palindromic obstructions.

Concretely: for genus 5, the hyperelliptic curve $y^2 = x^{11} + ax^9 + bx^7 + cx^5 + dx^3 + x$ has $\dim V_+ = 5$, and $\bigwedge^5(V_+)$ lives in $H^5(\mathrm{Jac}, \mathbb{Q})$ (odd cohomology). The Hodge Conjecture for odd cohomology of abelian varieties is known, so the real frontier is $g = 6$ (even cohomology, no hyper-Kähler structure, Kani–Rosen splitting potentially available on palindromic loci).

## 10.3 The conservation conjecture

Paper 90 conjectured that the Hodge Conjecture is *conservative* over BISH: its BISH-consequences are provable in BISH alone. Markman's theorem strengthens the evidence: the polynomial identities (Papers 84–89) that the Squeeze extracted are now known to be consequences of an unconditional truth. Conservation would require showing that these polynomial identities are provable without the Hodge Conjecture—a meta-mathematical question beyond the scope of this paper.

To make the conjecture precise: let $\varphi$ be a $\Pi^0_1$-sentence (a universal statement about integers, e.g., "this polynomial identity holds for all $n \in \mathbb{Z}$"). If HC $\vdash \varphi$, does BISH $\vdash \varphi$? The evidence

from 91 papers is consistent with a positive answer. Every $\Pi^0_1$-consequence of the Hodge Conjecture that the CRM program has identified has been verified in BISH by `ring` or `decide`. But a proof of conservation would require new proof-theoretic techniques—specifically, a cut-elimination procedure for the CLASS content of the cycle class map.

### 10.4 Relationship to Floccari–Fu

Floccari–Fu [4] prove HC for Weil fourfolds of discriminant 1 via singular OG6-varieties rather than secant sheaves; Markman's proof covers all discriminants. A CRM audit of their proof would likely yield a similar CLASS profile: the key classical input is the existence of a singular hyper-Kähler resolution (existence of birational model, Zorn), plus properness of the moduli space of sheaves on the singular variety. The BISH content (numerical invariants, lattice computations) would also be comparable.

The two proofs use different CLASS axioms to reach the same conclusion: Markman via Orlov + Schoen + analytic deformation; Floccari–Fu via singular birational geometry + moduli properness. This "axiom trading" is precisely what the CRM audit detects and quantifies.

### 10.5 The asymmetric offloading pipeline

The palindromic cycle construction demonstrates the mature form of the CAS-to-Lean pipeline first introduced in Paper 77. The architecture has three layers:

1. **Symbolic computation** (Python/SymPy): polynomial algebra, factorization, substitution. Trusted for computation, not for proof. The script is deterministic and reproducible.
2. **Code emission**: the CAS writes Lean 4 source code containing `theorem` declarations with `by ring` or `by decide` proof terms. The emitted code is human-readable and can be audited independently.
3. **Kernel verification** (Lean 4): the type checker verifies each theorem independently of the CAS. If the CAS emits an incorrect identity, `ring` will fail to close the goal.

This pipeline has been applied across the generative phase (Papers 77–78, 80–86, 88–89, 91), scaling from single-variable polynomials to 3-parameter families over $\mathbb{Z}[a, b, c]$. The key insight is that CAS tools are *oracles*, not provers: they suggest polynomial identities that the Lean kernel then certifies. The oracle is untrusted; only the kernel's verification counts.

For Paper 91, the CAS performed 9 verifications in the symbolic algebra of $\mathbb{Q}(a, b)[u]$ and emitted 177 lines of Lean with 12 `ring`-verified theorems. Total CAS runtime: $< 1$ second. Lean build time: $\sim$3 minutes (dominated by Mathlib compilation).

## 11 Conclusion

This paper performs the first CRM audit of a resolution of a sub-case of a Clay Millennium Problem. Four results:

(i) Markman's proof decomposes into 4 BISH + 5 CLASS components, with the 5 CLASS nodes identified precisely (Theorem A).

(ii) At the respective decomposition granularities, the CRMLint Squeeze (90% BISH, conditional) isolates classical content more sharply than Markman's approach (44% BISH, unconditional); the gap narrows to $\sim$67% vs. 90% at matched granularity (Remark 4.1), but the qualitative conclusion persists (Theorem B).

(iii) VHC is shown to be consistent (not refuted) for the genus-4 hyperelliptic Weil family: the Squeeze's conditional prediction is independently confirmed by Markman's unconditional result (Theorem C).

(iv) The cycle class map is the irreducible CLASS boundary; 20/0 is logically impossible; 19/1 is the best achievable (Theorem D).

On the palindromic locus, asymmetric offloading constructs the explicit algebraic cycle correspondence via Kani–Rosen genus-2 reduction, confirmed by 177 lines of CAS-emitted, Lean-verified polynomial identities.

Markman proved the cycle exists. CRM mapped its logical skeleton. The BISH/CLASS boundary is not eliminated; it is translated. The Hodge Horizon (WLPO, Paper 87) is robust under both the Squeeze and the derived-category approach.

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